1 **(a)** **(a)** $8000(1+\frac{p}{100})^3 = 27000$ **[M1]**

$1+\frac{p}{100} = \sqrt[3]{\frac{27000}{8000}} = \frac{3}{2}$ **[M1]**

50 **[A1]**

(b) $27000 \times (1.5)^2$ or $8000 \times (1.5)^5$ **[M1]**

60 750 **[A1]**

[Total Marks: 5]

2 $450 \times \left(1+\frac{6.5}{100}\right)^{12}$ **[M1]**

450×1.065^{12} **[M1]**

958.01 (accept 958) **[A1]**

[Total Marks: 3]

3 $32000 \times (1-0.12)^5$ oe **[M2]**

For 32000×0.88^n where $n > 1$, award **[M1]**

16887 **[A1]**

[Total Marks: 3]

4 **(a)** **(a)** $85000(1+\frac{r}{100})^{12} = 112000$ **[M1]**

$1+\frac{r}{100} = \left(\frac{112}{85}\right)^{\frac{1}{12}}$ **[M1]**

2.33 (from 2.327...) **[A1]**

(b) $85000 \times \left(1+\frac{2.327...}{100}\right)^{20}$ or $112000 \times \left(1+\frac{2.327...}{100}\right)^{8}$ **[M1]**

135 000 **[A1]**

[Total Marks: 5]

5 **(a)** $28\,500 \times \left(1-\frac{12}{100}\right)^5$ **[M1]**

$28\,500 \times 0.88^5$ **[M1]**

15 040 **[A1]**

(b) $35\,000 \times \left(1-\frac{r}{100}\right)^4 = 15\,200$ **[M1]**

$\left(1-\frac{r}{100}\right)^4 = \frac{152}{350}$ **[M1]**

$1-\frac{r}{100} = \sqrt[4]{\frac{152}{350}} = 0.8123\ldots$ **[M1]**

18.8 (accept 18.76 to 18.77) **[A1]**

[Total Marks: 7]

6 (a) $2500 \times \left(1 + \frac{r}{100}\right)^6 = 9100$ **[M1]**

$\left(1 + \frac{r}{100}\right)^6 = 3.64$ **[M1]**

$1 + \frac{r}{100} = 3.64^{\frac{1}{6}} = 1.2403\ldots$ **[M1]**

24.0 (accept $24.03\ldots$) **[A1]**

(b) 2500×1.2403^{10} or 9100×1.2403^4 **[M1]**

21 500 **[A1]**

(c) $2500 \times 1.2403^n > 50\,000 \implies 1.2403^n > 20$ **[M1]**

$n \times \log(1.2403) > \log(20) \implies n > 13.91\ldots$ (or equivalent trial and error) **[M1]**

14 **[A1]**

[Total Marks: 9]